

Rope Strand Dynamics: the Strand-Level Campaign

Phase 1 (FND-STRAND-001): the substrate -- inextensible elastic curves with the derived finite contact energetics; circle relaxation; dynamical linking conservation with finite standoff on a true Hopf pair; unlinked control. Phases 2-4: framed transport (FND-KIN-001 fidelity), the coverage threshold (FND-MATTER-004 fidelity), and the interaction model (carrying the detection law and the measurement wall, per QB-022). This anchor grows with the campaign.